

Memory ML, DL, NLP Course

①

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SEC-36

Gradient Boosting

Lec 1:- Gradient Boosting
Regression

Gradient Boosting Algorithm:

- ① Regression
- ② Classification

Regression Dataset

Ex	Degree	Salary
2	BE	50K
3	Masters	70K
5	Masters	80K
6	PHD	100K

Steps:

(1) Create a Base Model

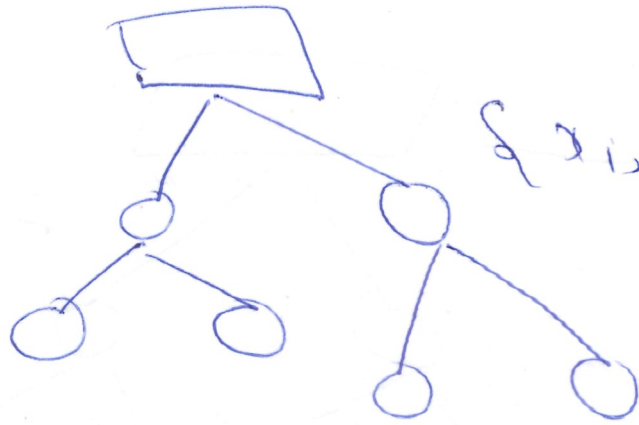
$$\boxed{75} \quad \text{Average} = \frac{50 + 70 + 80 + 100}{4} = 75$$

(2) Compute residuals, Error. etc.

Ex	Degree	Salary	$\hat{y}$	$(y_i - \hat{y}_i)$
2	BE	50K	75K	-25K
3	Masters	70K	75K	-5K
5	Masters	80K	75K	5K
6	PHD	100K	75K	25K
		$\bar{R}^2$		
		-23		
		-3		
		3		
		20		

(3) Construct a Regression Tree.
 Tree consider inputs
 x_i and output R_i

75



3

family

predicted $O(P) \Rightarrow 75 + (-23)$
 $\Rightarrow 52$
 $= 52$

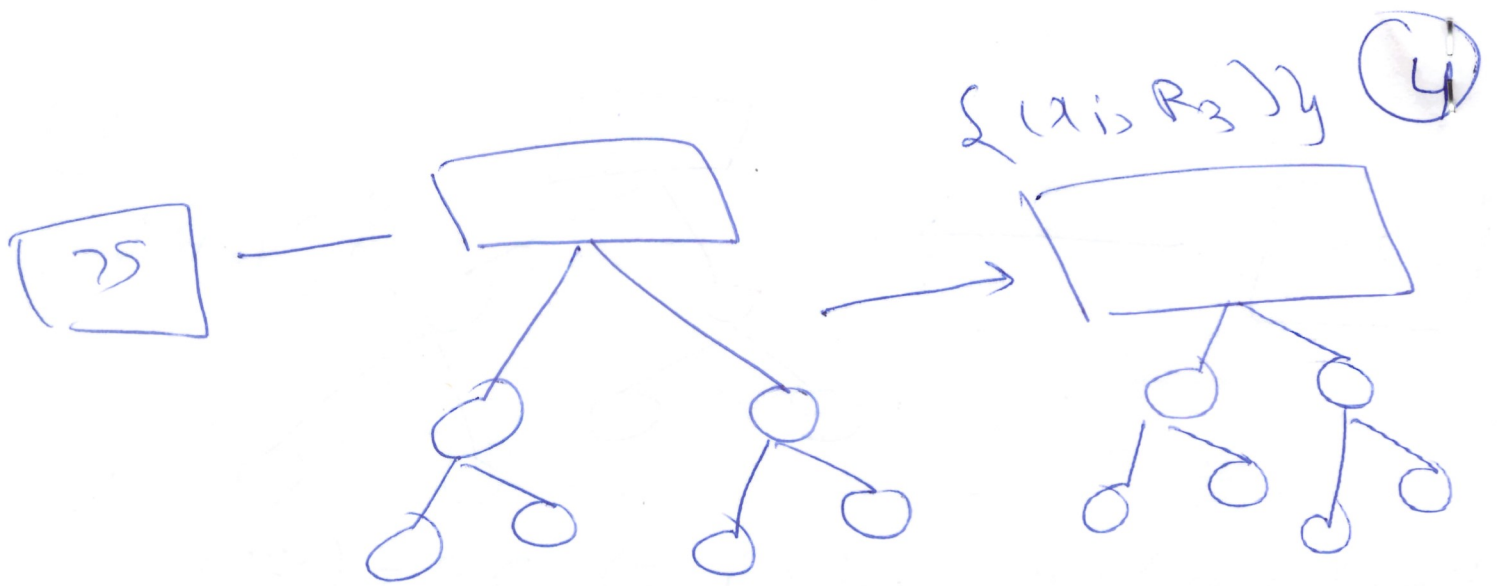
overfitting

predicted $O(P) \Rightarrow 75 + \alpha$

$\alpha = \text{learning rate}$
 $\alpha = 0.1$

predicted $O(P) \Rightarrow 75 + \alpha(-23)$
 $= 75 + (0.1)(-23)$
 $= 75 - 2.3$
 $= 72.7$

$= 75 + 0.1(-?)$
 $= 75 - 0.3$
 $= 74.7$



$$F(x) = h_0(x) + \alpha_1 (h_1(x))$$

$$F(x) = h_0(x) + \alpha_1 (h_1(x)) + \alpha_2 (h_2(x)) +$$

$$\alpha_3 (h_3(x)) + \dots + \alpha_n (h_n(x))$$

Learning Rate $\alpha = 0.1$

$$F(x) = \sum_{i=0}^n \alpha_i h_i(x)$$

Final function of Gradient Boosting